Annex II. Togo: Joint World Bank-IMF Debt Sustainability Analysis

The debt sustainability analysis (DSA) highlights that Togo is in debt distress, underscoring the importance of comprehensive debt relief.[18] *Under the external DSA baseline, Togo's NPV-based indicators are projected to decline over time but they breach the relevant indicative thresholds over a prolonged period and are vulnerable to a number of shocks. Under an alternative scenario, which assumes full delivery of HIPC, MDRI and beyond-HIPC debt relief at completion point, Togo's external debt indicators exhibit a substantial improvement. Including Togo's large domestic public debt only reinforces the conclusions of the external DSA.*

1. **This DSA is based on the Debt Sustainability Framework (DSF) for Low-Income Countries.** The DSA presents the projected path of Togo's external and public sector debt burden indicators, and assesses the country's risk of external debt distress. Methodologically, the DSA differs from the Debt Relief Analysis (DRA) in the decision point document in that it compares the evolution over the projection period of debt-burden indicators against policy-dependent indicative thresholds. In contrast, under the DRA, the historical debt burden indicators are compared to uniform thresholds in order to calculate the amount of HIPC debt relief that Togo qualifies for in the context of the HIPC Initiative.[19]

2. **The last DSA for Togo was prepared in 2007 and assessed Togo as being in debt distress.**[20] At that time, Togo was accumulating arrears to several multilaterals and several debt burden indicators showed substantial breaches of the respective indicative thresholds. The macroeconomic framework underpinning the 2007 DSA baseline assumed weak economic performance and continued arrears accumulation, resulting in a projected worsening of the debt burden indicators. Compared to 2007 DSA results, the analysis below finds an improved debt dynamics as a consequence of projected improvements in the macroeconomic policy framework, large arrears clearance operations, and HIPC Initiative interim debt relief.

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[18] This DSA has been prepared jointly by the World Bank and Fund staffs using the Debt Sustainability Framework (DSF) for Low Income Countries (see "Applying the Debt Sustainability Framework for Low-Income Countries Post Debt Relief", (IDA/SecM2006-0564, 8/11/06). Togo's quality of policies and institutions, as measured by the average World Bank's Country Policy and Institutional Assessment (CPIA) for the period 2005-2007 (2.5), places it as a "weak performer." The corresponding indicative thresholds for the external debt indicators are 30 percent for the NPV of debt-to-GDP ratio, 100 percent for the NPV of debt-to-exports ratio, 200 percent for NPV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio and 25 percent for the debt service-to-revenue ratio.

[19] In addition, the results of the LIC DSA differ from the HIPC DRA because of two other methodological differences related to the definition of: (i) discount rates; and (ii) exchange rates.

[20] Togo: 2007 Article IV Consultation and First Staff Review of the Staff-Monitored Program - Staff Report; Public Information Notice on the Executive Board Discussion; and Statement by the Executive Director for Togo.

A. Baseline Assumptions

3. **The baseline scenario relies on the same long-run macroeconomic framework as the DRA and is consistent with the program projections under the PRGF arrangement.** Average annual real GDP growth is projected to pick up over the projection period, from an average of 2 percent in 2004–2008 to 3.6 percent in 2009–2027. This pattern reflects the projected increase of donor-financed public investment, improved business confidence, growing regional trade, and a rebound in cotton and phosphate production. Over the medium term, growth should be sustained by higher foreign direct investment (e.g. in banking, telecom and phosphate sectors, and the port), improved financial intermediation, and upgrades to public infrastructure, especially electricity, which would alleviate bottlenecks that previously stymied private sector growth.

4. **Macroeconomic stability is expected to be supported by a prudent fiscal policy.** A domestic primary surplus of about 1 percent of GDP is assumed, which, combined with comprehensive external debt relief and higher foreign aid, would result in a broadly sustainable fiscal and debt position that avoids renewed arrears. The authorities plan to achieve the required fiscal adjustment by gradually raising revenues to about 18 percent over the medium run while holding domestic spending broadly constant in relation to GDP. Spending in priority areas (health, education, infrastructure, and restructuring of state-owned banks and enterprises) is to be increased significantly by (i) reallocating domestic spending away from non priority sectors, particularly untargeted subsidies and transfers, (ii) bringing external assistance (it was US$12 per capita in the period 2004-06, per the World Development Indicators database) to the regional average (US$43 per capita), and (iii) using resources to be freed up by HIPC and MDRI debt relief.

5. **Despite the global price shocks and the challenging external conditions, inflation is expected to be contained over the medium term, anchored in the WAEMU currency union and fiscal prudence.** Nonetheless, inflation can be erratic in case of unfavorable weather conditions (droughts or the recent heavy floods) or further volatility in world food and oil prices.

6. **Under the assumption that FDI and donor flows continue to be robust over the medium term, the external position is projected to remain manageable**. Togo's current account deficit is not expected to decline much over the medium term: imports are projected to increase as foreign aid is absorbed and FDI increases. Exports are expected to pick up, largely on account of higher phosphate exports, but sustained export growth will require enhancing competitiveness through reforms to improve the business environment. External financing is expected to come primarily from debt relief and higher FDI flows. Project grants are projected to increase from just under 2 percent of GDP in 2007 to a long-run average of 4¼ percent of GDP, whereas concessional project loans are projected to reach an average of 1½ percent of GDP.

7. **The baseline scenario reflects the multilateral arrears clearance operations completed in 2008 and assumes full delivery of traditional debt relief as well as interim HIPC assistance.** [21] Consistent with the DSF guidelines, the baseline does not reflect the delivery of HIPC, MDRI and bilateral or multilateral beyond-HIPC assistance after the completion point.[22] The evolution of Togo's debt indicators reflecting the full impact of debt relief under the HIPC Initiative, MDRI and bilateral or multilateral beyond-HIPC assistance is presented in a country-specific alternative scenario.

B. Debt Sustainability Analysis

External Debt Sustainability

8. **Togo's external debt indicators under the baseline remain above the relevant indicative thresholds over a prolonged period** (Table 1a, Figure 1)**.** The NPV of external debt declines from 71.5 percent of GDP in 2007 to 42.9 percent of GDP in 2008 and remains above the 30 percent indicative threshold until 2015. The NPV of external debt-to-revenue ratio moves below the 200 percent indicative threshold by 2013. Conversely, debt service ratios remain below the respective indicative threshold over the whole projection period, reflecting the high degree of concessionality of Togo's debt.

9. **The sensitivity analysis indicates that Togo's external debt outlook under the baseline is subject to considerable vulnerabilities** (Table 1b, Figure 1). Togo's debt indicators under the "historical scenario" are less favorable than in the baseline, illustrating that the debt dynamics in the baseline are contingent upon an improved macroeconomic performance compared to Togo's relatively weak historical performance. In addition, the indicators deteriorate significantly under a variety of temporary shocks, in particular the one on non-debt external financing. Such shocks result in the NPV-based indicators breaching the indicative thresholds over a long period. Conversely, since Togo's new borrowing is assumed to be on highly concessional terms, the shocks affect the debt service indicators to a lesser extent. The debt service indicators remain well below the respective threshold, except for the most extreme (composite) shock, in which case the debt service to revenue ratio would cross the relevant indicative threshold for several years.

10. **Full delivery of debt relief through HIPC Initiative, MDRI and possible bilateral and multilateral beyond-HIPC assistance, would significantly improve Togo's external debt outlook**. As shown in the alternative, country-specific scenario, reaching the completion point and the resulting full delivery of the above assistance would reduce all

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[21] Arrears to IDA and AfDB have been cleared in 2008 and Togo has agreed on modalities for the clearance of arrears with EIB, EU and IFAD. Togo has not yet reached arrears clearance with the remaining multilateral creditors.

[22] See "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (www.imf.org and IDA/SECM2007/0226, 03/05/2007).

external debt indicators significantly below the relevant indicative thresholds. Similar results were found in an alternative scenario of the 2007 DSA that assumed an improved macroeconomic outlook and debt relief under HIPC and MDRI.

Public Sector Debt Sustainability

11. **Including Togo's large domestic public debt in the analysis underscores the vulnerability of the baseline scenario** (Table 2a, Figure 2)**.** Togo's domestic debt burden is comparatively large, reflecting years of weak fiscal management and domestic arrears accumulation, as well as the need to recapitalize ailing banks. The NPV of total public debt is projected to remain relatively high over the next five years, at more than 50 percent of GDP and more than 200 percent of revenues, respectively. Given the assumed improvement in the macroeconomic outlook and the projected high degree of concessionality of financing in the baseline scenario, debt ratios would nevertheless fall steadily over the long run. However, the evolution of the debt indicators would be sensitive to a variety of shocks, which would increase debt level and debt service over the long run.

12. **Total public debt dynamics are particularly vulnerable to growth shocks, but also to real exchange rate depreciation.** (Table 2b, Figure 2). Full delivery of HIPC relief, MDRI and beyond-HIPC relief at the assumed completion point reduces these vulnerabilities, as shown in the alternative scenario. All three NPV-based indicators would be substantially lower than under the baseline and would decline further over the projection period.

C. Conclusion

13. **The DSA shows that Togo remains in debt distress, despite the large reduction in debt achieved through traditional debt relief mechanisms, expected delivery of HIPC interim assistance and arrears clearance operations in 2008.** Under the baseline, the NPV-based indicators remain well above its indicative thresholds for most of the projection period, underscoring the importance for additional debt relief. Debt relief under the HIPC Initiative, MDRI and beyond-HIPC assistance significantly improves Togo's external debt outlook. These results are consistent with those of the 2007 DSA.

14. **Alternative scenarios and bound tests highlight the vulnerability of Togo's external debt outlook.** Achieving a robust external debt outlook will depend on a sustained pick-up of real GDP growth, solid export and revenue performance, as well as prudent debt management. The inclusion of Togo's large domestic debt in the analysis reinforces the conclusions of the external DSA and underscores the vulnerability of Togo's debt prospects.

Table 1a. Togo: External Debt Sustainability Framework, Baseline Scenario, 2005-2028 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2005	2006	2007			2008	2009	2010	2011	2012	2013	2008-2013 Average	2018	2028	2014-2028 Average
External debt (nominal) 1/	**77.1**	**84.7**	**88.2**			**59.9**	**58.6**	**57.0**	**54.0**	**50.7**	**47.3**		**33.9**	**25.0**	
o/w public and publicly guaranteed (PPG)	77.1	84.7	88.2			59.5	57.9	56.1	52.9	49.4	45.9		31.8	22.8	
Change in external debt	-16.2	7.6	3.5			-28.4	-1.3	-1.6	-3.0	-3.3	-3.3		-1.8	-1.0	
Identified net debt-creating flows	-4.7	-0.7	-5.7			7.8	3.7	0.4	-0.8	-0.9	-1.0		-1.1	-1.3	
Non-interest current account deficit	**3.3**	**4.7**	**4.5**	**4.1**	**2.2**	**10.4**	**8.8**	**6.7**	**4.4**	**4.1**	**4.0**		**3.6**	**3.5**	3.6
Deficit in balance of goods and services	16.9	19.5	20.5			27.0	26.9	25.0	23.3	22.6	22.0		20.8	21.4	
Exports	40.3	42.3	41.9			40.4	43.6	44.9	45.5	46.1	46.7		47.9	49.3	
Imports	57.2	61.8	62.5			67.5	70.6	70.0	68.9	68.8	68.8		68.8	70.7	
Net current transfers (negative = inflow)	-12.4	-14.7	-15.9	-10.8	3.3	-16.8	-18.4	-18.5	-18.1	-17.7	-17.3		-16.9	-17.4	-16.9
o/w official	-1.2	-1.4	-1.7			-1.8	-3.8	-4.3	-4.3	-4.3	-4.3		-4.3	-4.4	
Other current account flows (negative = net inflow)	-1.2	-0.2	-0.2			0.2	0.2	0.2	-0.8	-0.9	-0.8		-0.4	-0.5	
Net FDI (negative = inflow)	**-2.2**	**-3.4**	**-2.5**	**-2.9**	**1.1**	**-2.5**	**-3.8**	**-4.5**	**-4.0**	**-4.0**	**-4.0**		**-4.0**	**-4.1**	-4.0
Endogenous debt dynamics 2/	**-5.8**	**-1.9**	**-7.7**			**-0.1**	**-1.3**	**-1.8**	**-1.2**	**-1.0**	**-1.0**		**-0.6**	**-0.6**	
Contribution from nominal interest rate	2.0	1.9	1.9			0.4	0.4	0.4	1.0	1.0	0.9		0.6	0.3	
Contribution from real GDP growth	-1.0	-2.9	-1.5			-0.6	-1.7	-2.2	-2.1	-2.0	-1.9		-1.2	-1.0	
Contribution from price and exchange rate changes	-6.8	-0.9	-8.1			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-11.6**	**8.3**	**9.2**			**-36.2**	**-5.0**	**-2.0**	**-2.2**	**-2.4**	**-2.3**		**-0.8**	**0.3**	
o/w exceptional financing	0.0	0.0	0.0			-33.6	-3.5	-22.1	-0.6	-0.5	-0.6		-0.5	-0.7	
PV of external debt 4/	…	…	71.5			42.9	43.5	43.3	41.4	39.0	36.4		25.5	18.1	
In percent of exports	…	…	170.4			106.0	99.7	96.5	90.9	84.4	77.8		53.2	36.7	
PV of PPG external debt	**…**	**…**	**71.5**			**42.5**	**42.9**	**42.5**	**40.3**	**37.6**	**34.9**		**23.5**	**15.8**	
In percent of exports	**…**	**…**	**170.4**			**105.2**	**98.2**	**94.5**	**88.4**	**81.6**	**74.6**		**48.9**	**32.1**	
In percent of government revenues	**…**	**…**	**421.2**			**256.3**	**247.2**	**241.2**	**221.2**	**206.8**	**191.7**		**128.9**	**87.0**	
Debt service-to-exports ratio (in percent)	**34.9**	**29.0**	**26.9**			**5.0**	**3.8**	**3.5**	**5.8**	**6.7**	**7.3**		**5.1**	**3.2**	
PPG debt service-to-exports ratio (in percent)	**9.3**	**8.4**	**8.5**			**5.0**	**3.8**	**3.5**	**5.8**	**6.7**	**7.3**		**5.1**	**3.2**	
PPG debt service-to-revenue ratio (in percent)	**23.8**	**21.1**	**21.0**			**12.1**	**9.5**	**8.9**	**14.5**	**17.0**	**18.9**		**13.5**	**8.6**	
Total gross financing need (Billions of U.S. dollars)	0.3	0.3	0.3			0.3	0.2	0.1	0.1	0.2	0.2		0.2	0.3	
Non-interest current account deficit that stabilizes debt ratio	19.5	-2.9	0.9			38.8	10.1	8.3	7.4	7.4	7.3		5.4	4.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	1.2	3.9	1.9	1.1	2.5	0.8	3.0	4.0	4.0	4.0	4.0	3.3	3.5	4.0	3.6
GDP deflator in US dollar terms (change in percent)	7.8	1.2	10.6	3.7	8.4	18.0	2.2	2.4	2.4	2.4	2.0	4.9	2.0	2.9	2.1
Effective interest rate (percent) 5/	2.3	2.6	2.5	2.1	0.5	0.6	0.7	0.7	1.8	2.0	2.0	1.3	1.7	1.4	1.6
Growth of exports of G&S (US dollar terms, in percent)	6.6	10.3	11.8	10.2	12.6	14.6	13.6	9.7	8.0	7.9	7.5	10.2	5.6	10.0	6.1
Growth of imports of G&S (US dollar terms, in percent)	10.8	13.7	13.9	9.4	11.5	28.4	10.1	5.6	4.9	6.4	6.1	10.2	5.6	10.0	5.9
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	32.9	32.9	32.7	32.8	32.6	32.2	32.7	34.8	33.1	33.8
Government revenues (excluding grants, in percent of GDP)	15.7	16.9	17.0			16.6	17.3	17.6	18.2	18.2	18.2		18.2	18.2	18.2
Aid flows (in Billions of US dollars) 7/	0.0	0.0	0.0			0.1	0.2	0.2	0.2	0.2	0.2		0.3	0.6	
o/w Grants	0.0	0.0	0.0			0.1	0.1	0.2	0.2	0.2	0.2		0.2	0.4	
o/w Concessional loans	0.0	0.0	0.0			0.0	0.0	0.1	0.1	0.1	0.1		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			3.7	5.5	5.9	5.2	5.2	5.2		5.2	5.3	5.3
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			63.6	72.3	76.0	81.0	81.2	80.9		81.8	81.0	81.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	2.1	2.2	2.5			3.0	3.1	3.3	3.6	3.8	4.0		5.3	9.3	
Nominal dollar GDP growth	9.1	5.2	12.7			18.9	5.3	6.5	6.5	6.5	6.1	**8.3**	5.6	7.0	5.7
PV of PPG external debt (in Billions of US dollars)			1.8			1.3	1.3	1.4	1.4	1.4	1.4		1.2	1.5	
(PVt-PVt-1)/GDPt-1 (in percent)						-20.9	2.6	2.4	0.4	-0.2	-0.6	-2.7	-0.2	0.3	0.0

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b.Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008-2028
(In percent)

											Projections										
	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
PV of debt-to GDP ratio																					
Baseline	43	43	42	40	38	35	32	29	27	25	**23**	22	22	21	20	19	19	18	17	17	16
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2008-2028 1/	43	43	45	43	41	38	36	33	32	32	**32**	32	33	33	34	34	34	34	33	33	33
A2. New public sector loans on less favorable terms in 2008-2028 2	43	43	43	41	39	37	35	32	30	29	**28**	27	27	27	27	26	26	26	26	25	25
A3. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	43	43	12	12	12	12	12	12	11	11	**11**	11	11	12	12	12	13	13	13	13	13
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	43	45	47	44	41	38	35	32	29	28	**26**	25	24	23	22	21	21	20	19	18	17
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	43	48	56	53	50	47	44	41	39	36	**33**	31	29	27	25	23	21	20	19	18	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	43	46	49	46	43	40	37	33	31	29	**27**	26	25	24	23	22	22	21	20	19	18
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	43	53	63	61	58	55	51	48	46	43	**39**	36	33	30	28	25	22	21	20	19	18
B5. Combination of B1-B4 using one-half standard deviation shocks	43	58	79	76	72	68	65	61	58	54	**49**	45	41	37	34	30	27	24	23	22	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	43	61	60	57	53	49	45	41	38	35	**33**	32	30	29	28	27	26	25	24	24	22
PV of debt-to-exports ratio																					
Baseline	105	98	94	88	82	75	68	61	56	52	**49**	47	45	43	42	40	39	37	36	35	32
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2008-2028 1/	105	99	100	95	89	82	76	70	66	66	**66**	67	69	70	71	71	71	71	70	68	66
A2. New public sector loans on less favorable terms in 2008-2028 2	105	96	93	86	77	68	59	50	42	35	**29**	23	19	15	12	9	6	3	1	-1	-2
A3. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	105	98	27	27	27	26	25	24	23	23	**23**	23	24	25	25	26	26	27	27	28	27
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	105	98	94	88	82	75	68	61	56	52	**49**	47	45	43	42	40	39	37	36	35	32
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	105	127	162	153	143	133	123	113	106	100	**91**	85	79	73	68	63	58	54	51	49	45
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	105	98	94	88	82	75	68	61	56	52	**49**	47	45	43	42	40	39	37	36	35	32
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	105	122	141	134	125	117	109	101	95	89	**81**	75	69	63	57	52	47	43	41	39	36
B5. Combination of B1-B4 using one-half standard deviation shocks	105	138	178	169	159	149	139	129	123	115	**105**	95	87	79	71	64	57	52	49	46	42
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	105	98	94	88	82	75	68	61	56	52	**49**	47	45	43	42	40	39	37	36	35	32
PV of debt-to-revenue ratio																					
Baseline	256	247	241	221	207	192	176	160	147	137	**129**	123	118	114	110	106	103	99	95	91	87
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2008-2028 1/	256	249	245	227	215	203	190	176	167	160	**154**	151	149	147	146	144	143	141	140	139	137
A2. New public sector loans on less favorable terms in 2008-2028 2	256	241	237	215	195	175	153	131	110	92	**75**	61	50	40	31	22	15	9	3	-1	-5
A3. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	256	247	69	68	68	67	66	63	61	60	**60**	61	63	65	67	68	70	71	72	74	74
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	256	258	265	243	228	211	193	176	162	151	**142**	135	130	126	122	117	113	109	104	100	96
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	256	276	317	293	277	261	244	227	213	200	**184**	170	159	148	137	127	116	108	103	99	93
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	256	265	278	255	238	221	202	184	169	158	**149**	142	136	132	127	123	118	114	109	105	100
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	256	307	361	334	318	301	283	265	251	235	**214**	197	181	166	151	137	123	113	108	103	96
B5. Combination of B1-B4 using one-half standard deviation shocks	256	335	447	415	396	376	355	334	318	299	**271**	247	225	205	185	166	147	134	127	120	112
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	256	350	341	313	293	271	249	226	208	195	**182**	174	167	162	156	151	145	140	134	129	123

Table 1b (continued).Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008-2028

(In percent)

Debt service-to-exports ratio

Baseline	5	4	3	6	7	7	8	8	7	5	**5**	4	4	3	3	4	4	4	4	3	3
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2008-2028 1/	5	4	3	6	7	8	8	8	7	6	**5**	4	4	4	5	5	5	6	7	7	7
A2. New public sector loans on less favorable terms in 2008-2028 2	5	4	4	6	7	8	8	7	6	5	**5**	4	4	4	4	4	4	4	4	4	4
A3. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	5	4	3	1	2	2	3	3	3	2	**2**	2	1	1	1	1	2	2	2	2	2
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	5	4	3	6	7	7	8	8	7	5	**5**	4	4	3	3	4	4	4	4	3	3
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	5	4	5	9	10	11	11	11	10	9	**11**	10	9	8	8	8	8	7	5	5	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	5	4	3	6	7	7	8	8	7	5	**5**	4	4	3	3	4	4	4	4	3	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	5	4	4	7	8	9	9	9	8	9	**10**	9	8	8	8	8	7	6	4	4	4
B5. Combination of B1-B4 using one-half standard deviation shocks	5	4	5	9	10	10	11	11	9	11	**14**	12	11	11	10	10	10	8	5	5	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	5	4	3	6	7	7	8	8	7	5	**5**	4	4	3	3	4	4	4	4	3	3

Debt service-to-revenue ratio

Baseline	12	9	9	15	17	19	20	20	17	14	**13**	11	10	9	9	9	9	9	9	9	9
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2008-2028 1/	12	9	8	14	17	20	21	22	19	15	**13**	12	11	11	12	13	14	16	18	18	18
A2. New public sector loans on less favorable terms in 2008-2028 2	12	9	9	15	18	20	20	20	16	13	**13**	11	11	11	11	11	11	11	11	11	10
A3. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	12	9	9	4	5	6	7	8	7	6	**6**	5	4	3	4	4	4	4	4	4	4
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2009-2010	12	10	10	16	19	21	22	22	19	16	**15**	12	11	10	10	10	10	10	10	10	9
B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/	12	9	10	17	19	21	22	22	19	19	**22**	20	18	17	17	16	16	14	11	10	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010	12	10	10	17	20	22	23	23	20	17	**16**	13	11	11	11	11	11	11	11	11	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/	12	9	11	18	20	22	23	23	21	23	**27**	24	22	21	21	20	20	16	11	11	10
B5. Combination of B1-B4 using one-half standard deviation shocks	12	10	13	22	24	26	27	28	24	28	**36**	32	29	28	27	26	26	20	14	13	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/	12	13	13	21	24	27	28	29	25	20	**19**	16	14	13	13	13	13	13	13	13	12
Memorandum item:																					
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	20	20	20	20	20	20	20	20	20	20	**20**	20	20	20	20	20	20	20	20	20	20

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Togo:Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-2028 1/
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2005	2006	2007	Average	Standard Deviation	2008	2009	2010	2011	2012	2013	2008-13 Average	2018	2028	2014-28 Average
Public sector debt 1/	101.4	106.1	106.8			80.2	83.6	73.9	67.5	61.9	56.7		36.5	25.1	
o/w foreign-currency denominated	80.9	80.5	82.1			59.0	57.8	55.9	52.8	49.3	45.8		31.7	22.7	
Change in public sector debt	14.9	4.8	0.7			-26.6	3.4	-9.7	-6.4	-5.5	-5.2		-2.8	-1.1	
Identified debt-creating flows	...	-8.6	-10.8			-29.8	-3.5	-23.9	-3.0	-2.8	-2.4		-1.4	-1.3	
Primary deficit	1.5	1.9	-0.2	-0.1	2.3	1.3	1.2	0.2	-0.8	-0.7	-0.6	0.1	0.1	0.2	0.1
Revenue and grants	16.9	18.3	18.7			19.2	21.8	22.6	22.8	22.8	22.8		22.8	22.9	
of which: grants	1.2	1.4	1.7			2.7	4.5	5.0	4.6	4.6	4.6		4.6	4.7	
Primary (noninterest) expenditure	18.4	20.2	18.5			20.6	23.1	22.8	22.0	22.1	22.2		22.9	23.1	
Automatic debt dynamics	...	-10.4	-10.6			-8.7	-2.5	-2.8	-2.2	-2.1	-1.8		-1.5	-1.6	
Contribution from interest rate/growth differential	...	-2.3	-2.6			-4.4	-2.3	-2.6	-2.0	-1.9	-1.8		-1.5	-1.5	
of which: contribution from average real interest rate	...	1.6	-0.6			-3.6	0.1	0.6	0.9	0.7	0.6		-0.2	-0.5	
of which: contribution from real GDP growth	-1.0	-3.8	-2.0			-0.8	-2.3	-3.2	-2.8	-2.6	-2.4		-1.3	-1.0	
Contribution from real exchange rate depreciation	...	-8.1	-8.0			-4.3	-0.2	-0.2	-0.2	-0.2	0.0		...	...	
Other identified debt-creating flows	0.0	-0.1	0.0			-22.4	-2.2	-21.3	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			-4.1	-2.2	-2.2	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			-18.3	0.0	-19.1	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	...	13.3	11.5			3.2	6.9	14.2	-3.3	-2.7	-2.8		-1.4	0.3	
Other Sustainability Indicators															
PV of public sector debt	20.5	25.7	91.2			63.3	68.5	60.3	54.9	50.3	45.8		28.1	18.2	
o/w foreign-currency denominated	0.0	0.0	66.6			42.2	42.8	42.4	40.2	37.6	34.8		23.4	15.8	
o/w external	...	...	66.6			42.2	42.8	42.4	40.2	37.6	34.8		23.4	15.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.8	13.5	11.1			4.0	3.9	3.8	3.9	4.1	4.1		3.9	4.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	121.2	140.4	488.5			329.1	313.7	267.2	240.6	220.7	200.7		123.6	79.3	
PV of public sector debt-to-revenue ratio (in percent)	130.3	151.9	537.5			381.7	395.0	342.5	301.7	276.1	251.6		154.7	99.8	
o/w external 3/	…	…	392.2			254.1	246.7	240.7	220.8	206.4	191.3		128.6	86.8	
Debt service-to-revenue and grants ratio (in percent) 4/	23.6	24.0	24.3			23.6	18.6	22.5	24.8	22.0	22.2		14.8	8.0	
Debt service-to-revenue ratio (in percent) 4/	25.4	26.0	26.7			27.4	23.4	28.8	31.1	27.5	27.8		18.5	10.1	
Primary deficit that stabilizes the debt-to-GDP ratio	-13.4	-2.8	-0.8			28.0	-2.1	9.9	5.6	4.8	4.6		2.9	1.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	1.2	3.9	1.9	1.1	2.5	0.8	3.0	4.0	4.0	4.0	4.0	3.3	3.5	4.0	3.6
Average nominal interest rate on forex debt (in percent)	2.3	2.6	2.5	2.1	0.5	0.6	0.7	0.7	1.8	2.1	2.0	1.3	1.8	1.5	1.7
Average real interest rate on domestic debt (in percent)	...	...	-0.6	...	...	-3.4	0.1	0.8	1.2	1.1	0.9	0.1	-0.4	-2.3	-0.9
Real exchange rate depreciation (in percent, + indicates depreciation)	13.1	-10.3	-10.2	-4.0	11.5	-5.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.7	0.3	1.3	1.5	5.3	4.8	1.2	2.0	2.0	2.0	2.0	2.3	2.0	2.9	2.1
Growth of real primary spending (deflated by GDP deflator, in percent)	0.2	0.1	-0.1	0.0	0.2	0.1	0.2	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	32.9	32.9	32.7	32.8	32.6	32.2	32.7	34.8	33.1	...

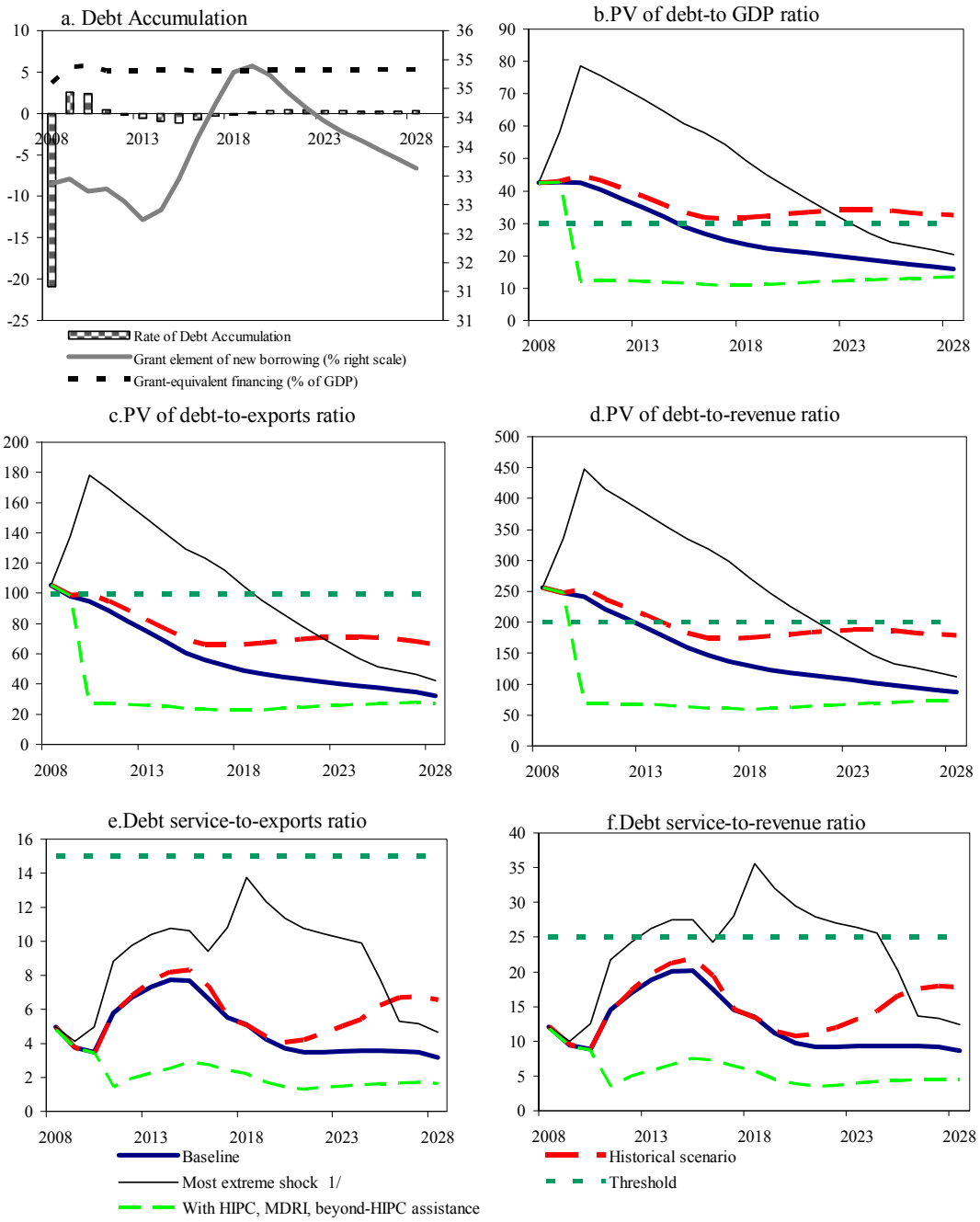
Sources: Country authorities; and Fund staff estimates and projections.
1/ Data covers gross central government debt.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Togo: Sensitivity Analysis for Key Indicators of Public Debt 2008-2028

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
PV of Debt-to-GDP Ratio								
Baseline	63	69	60	55	50	46	28	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	63	68	60	56	53	49	34	28
A2. Primary balance is unchanged from 2008	63	68	60	56	53	50	39	42
A3. Permanently lower GDP growth 1/	63	69	60	54	50	46	32	40
A4. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	63	69	30	27	25	23	16	16
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	63	72	67	62	58	55	44	52
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	63	69	62	56	50	46	28	21
B3. Combination of B1-B2 using one half standard deviation shocks	63	70	64	58	54	50	34	21
B4. One-time 30 percent real depreciation in 2009	63	86	76	69	64	59	40	32
B5. 10 percent of GDP increase in other debt-creating flows in 2009	63	78	69	62	56	50	29	18
PV of Debt-to-Revenue Ratio 2/								
Baseline	329	314	267	241	221	201	124	79
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	329	311	263	241	226	210	139	107
A2. Primary balance is unchanged from 2008	329	312	263	242	228	213	162	162
A3. Permanently lower GDP growth 1/	323	291	242	230	227	226	215	216
A4. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	329	314	133	119	110	101	69	69
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	329	326	292	268	252	236	189	222
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	329	317	273	243	222	200	123	92
B3. Combination of B1-B2 using one half standard deviation shocks	329	320	279	252	233	213	140	82
B4. One-time 30 percent real depreciation in 2009	329	394	337	304	281	258	175	140
B5. 10 percent of GDP increase in other debt-creating flows in 2008	80	93	82	74	67	61	37	25
Debt Service-to-Revenue Ratio 2/								
Baseline	24	19	22	25	22	22	15	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	19	23	23	23	24	20	9
A3. Permanently lower GDP growth 1/	24	19	23	25	23	24	20	24
A3. Permanently lower GDP growth 1/	24	19	34	30	36	36	39	38
A4. Alternative Scenario : full delivery of HIPC, MDRI and beyond HIPC debt relief	23	18	22	16	12	12	9	5
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	24	19	24	29	29	31	29	33
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	24	19	23	28	27	24	17	10
B3. Combination of B1-B2 using one half standard deviation shocks	5	4	5	6	6	6	5	2
B4. One-time 30 percent real depreciation in 2008	5	4	6	7	7	7	7	5
B5. 10 percent of GDP increase in other debt-creating flows in 2008	5	4	6	10	-5	7	1	0

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the length of the projection period.
2/ Revenues are defined inclusive of grants.

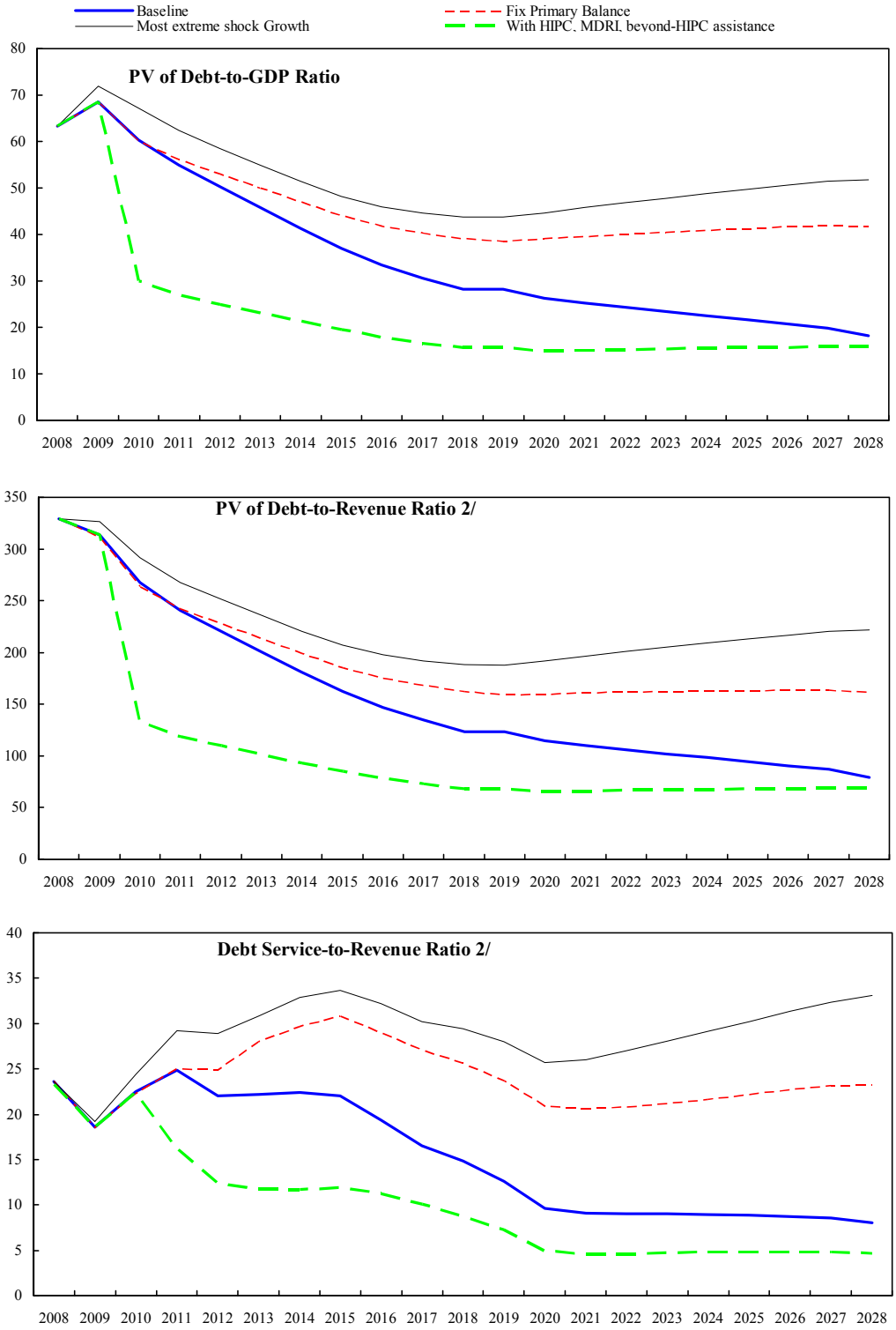
Figure 1. Togo: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2008-2028 1/



Source: Staff projections and simulations.

1/ The most extreme shock corresponds to the combination of B1-B4 using one-half standard deviations.

Figure 2.Togo: Indicators of Public Debt Under Alternative Scenarios, 2008-2028 1/



Sources: Country authorities; and Fund staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2018.
2/ Revenues are defined inclusive of grants.